Junyi Zhao

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Education

The University of Hong Kong | Hong Kong SAR 2026–2030
- **Ph.D.** in Earth and Planetary Sciences
- **Research Focus:** Lunar Science

University of Michigan, Ann Arbor | Ann Arbor, MI 2022–2026
- **Majors:** Astronomy & Astrophysics; Earth and Environmental Sciences; Interdisciplinary Physics
- **Minors:** Music; Oceanography

Honors & Awards

HKU Presidential PhD Scholarship (HKU-PS), The University of Hong Kong 2026-2030
Summer Undergraduate Research Fellowships (SURF), Caltech 2025
University Honors, University of Michigan 2023-2025

Publications

Zhao, J., Han, Y., et al. (in prep.) *Multiwavelength Analysis of the Vertical Structure of the AU Microscopii Debris Disk Using Joint JWST NIRCam and ALMA Modeling.*

Poster & Science Talks

Multi-wavelength Analysis of the AU Microscopii Debris Disk vertical structure
AAS 247 oral presentation | Phoenix, AZ Jan 2026

A joint model study of the scale height of debris disk AU-mic
Invited talk, KIAA Joint Group Meeting, Peking University | Beijing, CHN Dec 2025
SURF research presentation | Pasadena, CA Aug 2025

Searching for protoplanets orbiting MWC 758 using JWST/NIRCam
Undergraduate Research Symposium 2025 | Ann Arbor, MI May 2025

Identification and Kinematics of OBe stars in the LMC
Undergraduate Research Symposium 2024 | Ann Arbor, MI May 2024

The night sky – Constellation, star,and planets
Astro 101 observation section | Santa Barbara, CA May 2022

Research and lab Experience

Evaluating the YSES 3 Planetary-Mass Candidate with JWST/NIRCam Jan 2026 –
With: Aniket Sanghi, Prof. Beth Biller, Dr. Aarynn Carter
- This project uses JWST/NIRCam relative astrometry to test whether the wide-separation planetary-mass candidate in the YSES 3 system is a bound companion or a background object.

Constraining Grain-Size-Dependent Dynamics in AU Mic's Debris Disk Jun 2025 –
Caltech, CA | Adviser – Dr.Yinuo Han
- Built a unified Bayesian framework to jointly model JWST/NIRCam F444W scattered light and ALMA Band 6 visibilities under shared geometry to measure grain-size-dependent scale heights.
- Reduced JWST/NIRCam coronagraphy in SpaceKLIP; ran RDI using winnie;
- Fit ALMA data in the visibility domain with Galario using a disk model. Multiple radial profiles tested for model.
- Paper in prep; preparing to present results at the 2026 Winter AAS.

Searching for protoplanets orbiting MWC 758 using JWST/NIRCam Jul 2024 – May 2025
University of Michigan, MI | Adviser – Prof. Michael Meyer

- Aimed to check for the existence of protoplanet MWC758c using F200w and F430M JWST/NIRCam bands.
- Utilized Angular Differential Imaging (ADI) and Reference star differential imaging (RDI) using Pynpoint.
- Presented research findings at the Astronomy department's Undergraduate Research Symposium 2025.

Identification and Kinematics of OBe stars in the LMC Feb 2023 – May 2024
University of Michigan, MI | Adviser – Prof. Sally Oey

- Developed a photometric methodology for the selection of Oe and Be-type stars in the Large Magellanic Cloud.
- Designed and implemented a Python-based star selection algorithm, incorporating Aperture Photometry techniques.
- Analyzed Gaia mission data to accurately calculate the transverse velocities of Oe and Be- type stars within the Large Magellanic Cloud.
- Presented research findings at the Astronomy department's Undergraduate Research Symposium 2024.

Interferometric Data Analysis of Be stars May – Jul 2024
University of Michigan, MI | Adviser – Prof. John Monnier

- Processed interferometry data of scientific binary systems and Be stars using IDL (Interactive Data Language) to automate data reduction pipelines.
- Gained knowledge of interferometric techniques and their application in astronomical research.

High-Pressure Lab assistant Jun 2024
University of Michigan, MI | Adviser – Prof. Jie (Jacky) Li

- Utilized microscopy techniques to prepare laboratory consumables for high-pressure experimental procedures.
- Trained in laboratory protocols and safety guidelines for high-pressure experiments.

Outreach Talks

International collaboration in Chang'e program | Ann Arbor, MI Oct 2025
Mid-Autumn Festival U-M Astronomy Presentations 2025

The dark side of the moon–Chang'e program | Ann Arbor, MI Sep 2024
Mid-Autumn Festival U-M Astronomy Presentations 2024

Fresh samples from the Moon–Chang'e program | Ann Arbor, MI Sep 2023
Mid-Autumn Festival U-M Astronomy Presentations 2023

Science Communication & Public Outreach

Active member of the University of Michigan Student Astronomical Society (SAS) | Ann Arbor, MI
Guest presenter at the Detroit observatory | Ann Arbor, MI
Founder and President of the School Astronomy Club, Beijing No.4 High School | Beijing, China
Active instructor of the School Astronomy Club, Beijing No.4 High School | Beijing, China
Volunteer Planetarium Guide and Presenter at Beijing Planetarium. | Beijing, China
Lead meteor shower observation and photography group on a 3 day trip. | Hebei, China

Data Science Skills

Proficient:	Python, Markdown, DS9, Jupyter, ArcGIS Pro
Intermediate:	SQL, MESA-web, Cloud Computing, LaTeX
Familiar:	C++, HTML, Docker, MATLAB
Datasets:	JWST Near Infrared Camera (NIRCam) ALMA (Atacama Large Millimeter/submillimeter Array) GAIA DR3 CTIO (Curtis/Schmidt Telescope) CHARA (Center for High Angular Resolution Astronomy) Array
Packages:	pandas, numpy, matplotlib, astropy, scipy, seaborn, scikit-learn, SpaceKLIP, pynpoint, disc2radmc, winnie, galario, emcee, applefy, astroquery